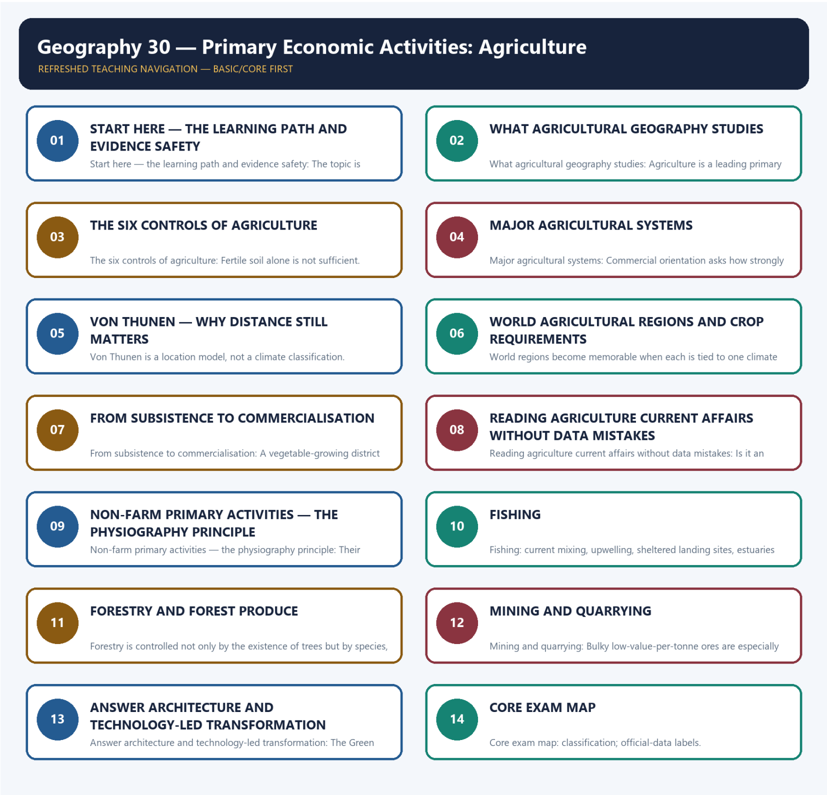
SOLVED PRACTICE WORKBOOK

Geography 30 - Primary Economic Activities: Agriculture - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Geography 30 - Primary Economic Activities: Agriculture

BASIC MCQS / REMEDIATION

MCQ 1 - Basic mastery check

What is the central question of agricultural geography?

- A. Where production occurs, why it occurs there and how its seasonal organisation changes
- B. Only how seeds are bred
- C. Only which crop has the highest output
- D. Only how farm machinery works

Answer: A. Where production occurs, why it occurs there and how its seasonal organisation changes

Why: Agricultural geography explains spatial organisation. Agronomy and machinery matter, but they do not replace the location question.

Remediation if missed: Return to the four core words: location, seasonality, organisation and change.

MCQ 2 - Basic mastery check

Which statement best explains why fertile soil alone does not create an agricultural region?

- A. Only government policy matters
- B. Relief always overrides every other factor
- C. Soil has no role in crop choice
- D. Climate, water, labour and market access must also align

Answer: D. Climate, water, labour and market access must also align

Why: Soil is one filter in a multi-control system. A crop still needs suitable heat, water, labour arrangements and access.

Remediation if missed: Rebuild the six-control table and test one Indian crop against every control.

MCQ 3 - Basic mastery check

Which feature most clearly distinguishes shifting cultivation from plantation agriculture?

- A. Both are necessarily mechanised
- B. Plantations always use long fallow
- C. Shifting cultivation rotates temporary plots and fallow; plantations are specialised commercial estates
- D. Both are urban-oriented systems

Answer: C. Shifting cultivation rotates temporary plots and fallow; plantations are specialised commercial estates

Why: The two systems differ in plot cycle, scale and market orientation. Plantation production is commercial and usually specialised.

Remediation if missed: Compare every farming system on climate, labour/scale and market orientation.

MCQ 4 - Basic mastery check

Which statement correctly describes mixed farming?

- A. It excludes animals
- B. It is identical to nomadic herding
- C. Crops and livestock are integrated so that outputs and inputs can support one another
- D. It must be a tropical plantation

Answer: C. Crops and livestock are integrated so that outputs and inputs can support one another

Why: Mixed farming integrates enterprises: residues may feed livestock and manure may return nutrients to fields.

Remediation if missed: Draw a two-way crop-residue/manure arrow rather than memorising a label.

MCQ 5 - Basic mastery check

Which statement best separates intensive farming from commercial farming?

- A. Intensive always means large farms
- B. Intensity concerns inputs or output per unit land; commercial orientation concerns production for markets
- C. The terms are climatic regions
- D. Commercial always means low labour use

Answer: B. Intensity concerns inputs or output per unit land; commercial orientation concerns production for markets

Why: The terms use different axes. A small wet-rice farm can be intensive, while a plantation can be strongly commercial.

Remediation if missed: Ask whether the word describes land-use intensity or the destination of output.

MCQ 6 - Basic mastery check

In Von Thunen's logic, what most strongly pulls a product closer to the market?

- A. High transport cost, perishability or rapid quality loss with distance
- B. The absence of consumer demand
- C. Uniform land quality by itself
- D. A lower delivery frequency

Answer: A. High transport cost, perishability or rapid quality loss with distance

Why: Distance imposes a larger penalty on bulky, frequently delivered or perishable goods, reducing the rent they can pay farther away.

Remediation if missed: Use the chain distance -> freight/decay -> lower net return -> location response.

MCQ 7 - Basic mastery check

What is the best description of cold-chain effects on Von Thunen's model?

- A. Cold chains convert the model into a climate classification
- B. Cold chains stretch feasible market distance but add cost and infrastructure dependence
- C. Cold chains make all farm locations equally profitable
- D. Cold chains restore the exact classical rings

Answer: B. Cold chains stretch feasible market distance but add cost and infrastructure dependence

Why: Refrigeration reduces decay and can move perishables farther, but logistics, energy and reliability still matter.

Remediation if missed: Do not use 'technology removes distance'; use 'technology changes the distance penalty.'

MCQ 8 - Basic mastery check

Which statement about Von Thunen is correct?

- A. It predicts one permanent crop map for every city
- B. It is a location model based on market distance, transport cost and land rent
- C. It ranks countries by agricultural output
- D. It classifies monsoon climates

Answer: B. It is a location model based on market distance, transport cost and land rent

Why: The model is an abstract mechanism, not a literal universal map. Its assumptions help isolate the role of distance.

Remediation if missed: Memorise the three anchors: market distance, transport cost and land rent.

MCQ 9 - Basic mastery check

Which pairing is most accurate?

- A. Mediterranean margins - tropical wet-rice monoculture
- B. Prairies - shifting cultivation
- C. Urban-industrial belts - nomadic herding
- D. Monsoon Asia - intensive wet-rice supported by summer rain, alluvial lowlands and dense labour

Answer: D. Monsoon Asia - intensive wet-rice supported by summer rain, alluvial lowlands and dense labour

Why: Monsoon Asian wet-rice combines seasonal water, suitable lowlands and high labour availability.

Remediation if missed: Pair each world region with climate, system and labour/market feature.

MCQ 10 - Basic mastery check

Which climate pattern supports classic Mediterranean agriculture?

- A. Winter rain and summer drought
- B. Permanent frost
- C. Rain in every month with no dry season
- D. Only summer monsoon rain

Answer: A. Winter rain and summer drought

Why: Winter rain supports winter crops while the dry sunny summer favours vines, olives and horticulture with water management.

Remediation if missed: Use the seasonal phrase 'winter rain, summer drought' before recalling crops.

MCQ 11 - Basic mastery check

Why is commercial grain farming associated with the Prairies, Pampas, Steppes and Downs?

- A. Vast temperate plains permit large fields and mechanisation
- B. They require dense manual transplanting
- C. These are humid tropical deltas
- D. They lack transport to markets

Answer: A. Vast temperate plains permit large fields and mechanisation

Why: Broad level plains and lower population density support large mechanised holdings and bulk grain movement.

Remediation if missed: Connect relief and holding scale to mechanisation instead of memorising region names alone.

MCQ 12 - Basic mastery check

Why do market gardening and dairy often occur near cities?

- A. They can grow only on black soil
- B. Perishability, frequent delivery and dense demand favour accessible locations
- C. They are always subsistence activities
- D. They require no consumers

Answer: B. Perishability, frequent delivery and dense demand favour accessible locations

Why: These goods lose value with delay and benefit from nearby demand, though cold chains can extend their range.

Remediation if missed: Apply Von Thunen's mechanism to one everyday product such as milk or leafy vegetables.

MCQ 13 - Basic mastery check

Which statement about cotton location is safest?

- A. Cotton is a cool wet highland crop
- B. Cotton grows only in deltaic new alluvium
- C. A long warm frost-free season and suitable drainage matter; cotton is not confined to black soil
- D. Cotton requires permanent waterlogging

Answer: C. A long warm frost-free season and suitable drainage matter; cotton is not confined to black soil

Why: Black soil is favourable in parts of India, but alluvial cotton belts show that crop location is a bundle of conditions.

Remediation if missed: Replace one-soil absolutism with climate + drainage + water + market.

MCQ 14 - Basic mastery check

Which combination best suits tea?

- A. Warm humid conditions, acidic well-drained slopes and abundant plucking labour
- B. Waterlogged lowlands with no drainage
- C. Arid plains and saline soil
- D. Cold steppe grasslands

Answer: A. Warm humid conditions, acidic well-drained slopes and abundant plucking labour

Why: Tea benefits from humidity and slope drainage; heavy rain does not mean stagnant water is desirable.

Remediation if missed: Remember the paired condition: high moisture but good drainage.

MCQ 15 - Basic mastery check

Which statement about millets is most accurate?

- A. They require permanent flooding
- B. They are nutritionally inferior by definition
- C. They are confined to humid deltas
- D. Many tolerate lower or variable rainfall and lighter soils, with important nutrition and resilience value

Answer: D. Many tolerate lower or variable rainfall and lighter soils, with important nutrition and resilience value

Why: Millets fit many dryland systems and the label 'coarse cereal' is a grain-size or category label, not a nutrition verdict.

Remediation if missed: Link millets to dryland suitability without claiming every millet has identical requirements.

MCQ 16 - Basic mastery check

Which sequence best describes commercialisation?

- A. Export first, then production
- B. Subsistence -> surplus -> specialisation -> processing/value-chain linkage
- C. Subsistence directly to processing without surplus
- D. Mechanisation alone, without markets

Answer: B. Subsistence -> surplus -> specialisation -> processing/value-chain linkage

Why: Commercialisation is a staged transition enabled by water, technology, roads, storage, finance and demand.

Remediation if missed: Rehearse both the sequence and the enabling conditions beneath it.

MCQ 17 - Basic mastery check

Which institution-role pairing is most defensible?

- A. Irrigation agencies have no effect on cultivation
- B. FAO directly procures every farmer's crop
- C. Cooperatives, mandis and farmer organisations can aggregate output and connect producers with markets
- D. Agriculture ministries determine local rainfall

Answer: C. Cooperatives, mandis and farmer organisations can aggregate output and connect producers with markets

Why: Market institutions organise exchange and bargaining, while water and policy institutions shape other parts of the system.

Remediation if missed: Separate water control, policy direction, market linkage and international comparison roles.

MCQ 18 - Basic mastery check

Which is the safest interpretation of a Kharif MSP announcement?

- A. It proves every farmer sold at MSP
- B. It measures household nutrition
- C. It is a final harvest estimate
- D. It is a season- and crop-specific price-policy decision, not proof of procurement coverage or final output

Answer: D. It is a season- and crop-specific price-policy decision, not proof of procurement coverage or final output

Why: Announcement, procurement and output are different stages and measures. The date and crop season must remain attached.

Remediation if missed: Use the firewall: announcement != purchase != production.

MCQ 19 - Basic mastery check

Which set contains only non-farm primary activities in this topic?

- A. Manufacturing, banking and transport
- B. Software, tourism and dairy processing
- C. Fishing, forestry/gathering, mining and quarrying
- D. Crop farming, retail and construction

Answer: C. Fishing, forestry/gathering, mining and quarrying

Why: These activities directly harvest or extract natural resources and exclude manufacturing.

Remediation if missed: Define primary activity first, then explicitly exclude factories and services.

MCQ 20 - Basic mastery check

Which statement best explains fishing geography?

- A. All coastlines offer identical potential
- B. Only distance from the national capital matters
- C. Shelf form, nutrients, currents/upwelling, shelter, inland waters and access jointly shape location
- D. Fish production is impossible inland

Answer: C. Shelf form, nutrients, currents/upwelling, shelter, inland waters and access jointly shape location

Why: Physiography creates marine and inland potential, while harbours, technology, regulation and markets shape realised production.

Remediation if missed: Build a mechanism for shelf, upwelling, estuary and reservoir rather than listing coasts.

MCQ 21 - Basic mastery check

Which is the best explanation of forestry location?

- A. Every forest is equally suited to commercial timber
- B. Non-timber forest produce has no livelihood role
- C. Relief has no effect on extraction cost
- D. Climate and altitude shape forest type, while slope and accessibility shape extraction

Answer: D. Climate and altitude shape forest type, while slope and accessibility shape extraction

Why: Forest composition and reachability matter together; remote or steep terrain can constrain use even when forests are extensive.

Remediation if missed: Use the sequence forest type -> access -> product -> livelihood.

MCQ 22 - Basic mastery check

Why can two mineral deposits of similar grade have different economic outcomes?

- A. All ores have the same value density
- B. Relief, transport, power, processing and rights can make one deposit accessible and the other unviable
- C. Geology becomes irrelevant after discovery
- D. Mining is independent of infrastructure

Answer: B. Relief, transport, power, processing and rights can make one deposit accessible and the other unviable

Why: Occurrence is geological, but viable extraction depends on moving material, obtaining inputs and managing institutions.

Remediation if missed: Separate 'deposit exists' from 'resource is economically usable.'

MCQ 23 - Basic mastery check

Why does a technology-led agricultural package spread unevenly?

- A. Assured water, level land, credit, inputs, extension and procurement outlets are spatially uneven
- B. Yields cannot respond to improved seed
- C. Technology has no relation to infrastructure
- D. Every region begins with identical water and credit access

Answer: A. Assured water, level land, credit, inputs, extension and procurement outlets are spatially uneven

Why: The package works through complementary preconditions. Regions that already possess them become early surplus cores.

Remediation if missed: Trace precondition filter -> adoption -> yield response -> regional divergence.

MCQ 24 - Basic mastery check

Which structure best answers a Geography Mains question?

- A. A long list without a thesis
- B. A conclusion unrelated to the directive
- C. Only a scheme list
- D. Claim -> named evidence -> spatial/causal analysis -> qualification -> reasoned verdict

Answer: D. Claim -> named evidence -> spatial/causal analysis -> qualification -> reasoned verdict

Why: The structure keeps evidence analytical and ensures the conclusion answers the directive rather than merely ending the response.

Remediation if missed: After every example, write one sentence beginning 'This shows...' and add one real limitation.

Basic remediation ladder

Error pattern	Repair action	Mastery test
Classification errors	Rebuild the common comparison axes	Explain two systems without using memorised region names
Location errors	Draw control -> mechanism -> location	Add one India example and one exception
Model errors	Separate assumptions, mechanism and modern modification	Apply Von Thunen to milk or vegetables
Data-status errors	Write stage, date, unit and measure beside the figure	Explain why announcement is not outcome
Non-farm errors	Start from physiography and resource occurrence	Cover fishing, forestry, mining and quarrying without manufacturing

PYQS AND ANSWER PRACTICE

Verified direct PYQs

UPSC Prelims 2019 Q21 - New World crops

Which group of plants was domesticated in the New World and introduced into the Old World? A Tobacco, cocoa and rubber; B Tobacco, cotton and rubber; C Cotton, coffee and sugarcane; D Rubber, coffee and wheat.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A. Confidence: high.

Tobacco, cacao and the rubber tree are New World domesticates. Coffee and wheat are Old World; cotton has multiple domestication centres, so options containing coffee/wheat or an ambiguous cotton bundle are eliminated.

Why this earns marks: source-status transparency plus option-by-option elimination.

UPSC Prelims 2020 Q87 - frost-free subtropical crop

A subtropical crop is injured by hard frost, needs at least 210 frost-free days, 50-100 cm rainfall and light well-drained moisture-retentive soil. Identify it: A Cotton; B Jute; C Sugarcane; D Tea.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A. Confidence: high.

The long frost-free season with moderate rainfall and a light, well-drained moisture-retentive soil is the classic cotton requirement bundle. Jute and tea require more humid conditions; sugarcane has a different water-duration profile.

Why this earns marks: source-status transparency plus option-by-option elimination.

UPSC Prelims 2022 Q62 - tea-producing states

Among Andhra Pradesh, Kerala, Himachal Pradesh and Tripura, how many are generally known as tea-producing States? A one; B two; C three; D all four.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high.

Kerala, Himachal Pradesh and Tripura have recognised tea sectors, and Andhra Pradesh has tea cultivation including the Araku/Eastern Ghats context. The missing local official key prevents upgrading the answer to official status.

Why this earns marks: source-status transparency plus option-by-option elimination.

UPSC Prelims 2023 Q4 - India-China agriculture comparison

Statements: India has more arable area than China; India's proportion of irrigated area is higher; India's average productivity per hectare is higher. How many are correct? A one; B two; C all three; D none.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B. Confidence: medium-high.

The first two comparative statements are defensible under the data vintage used by the question, while India's average agricultural productivity per hectare is not higher than China's. Cross-country definitions and reference years are the main caution.

Why this earns marks: source-status transparency plus option-by-option elimination.

UPSC Prelims 2025 Q24 - turmeric, official key conflict

Statements for 2022-23: India is the largest producer and exporter of turmeric; more than 30 varieties are grown; Maharashtra, Telangana, Karnataka and Tamil Nadu are major producers. Options: A I-II; B II-III; C I-III; D I-II-III.

OFFICIAL SET-A KEY: B (statements II and III only).

The local official UPSC key unambiguously records B. However, dated PIB material describes India as the world's largest producer, consumer and exporter and supports statements II and III. The package preserves the official key and flags the unresolved primary-source contradiction rather than inventing a reconciliation.

Why this earns marks: source-status transparency plus option-by-option elimination.

2018 GS-I Q15

Define the Blue Revolution and explain problems and strategies for pisciculture development in India.

Claim: Define fisheries/aquaculture expansion, distinguish capture from culture, map inland/coastal potential, and evaluate seed-feed-disease-water-market constraints.

- **Named evidence/example:** India's floodplains, reservoirs and ponds show that fish geography is not coastal only.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** PM Matsya Sampada-type value-chain logic: seed, feed, health, landing, cold chain and market.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** Coastal aquaculture illustrates export potential with disease, salinity and habitat risks.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.

Qualification: Do not reduce Blue Revolution to a slogan or a production number.

Verdict: A region-specific, biosecure and cold-chain-linked fisheries strategy can diversify rural income without externalising ecological costs.

Why this earns marks: directive fidelity, named evidence, analysis, qualification and a reasoned verdict.

2022 GS-I Q15

Describe the distribution of rubber-producing countries and indicate major environmental issues faced by them.

Claim: Locate humid tropical South-East and South Asia, parts of Africa and Latin America; explain heat-rainfall and plantation-processing geography.

- **Named evidence/example:** Thailand, Indonesia, Vietnam and Malaysia illustrate the South-East Asian core.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** India's classic belt is Kerala with suitable expansion pockets in the North-East.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** Plantation expansion can replace diverse forests, simplify habitats and create chemical/soil impacts.

- **Analysis:** The evidence proves a specific spatial or institutional mechanism.

Qualification: Natural-rubber distribution also reflects colonial planting, labour, disease and processing, not climate alone.

Verdict: Sustainable rubber requires yield improvement on existing land, diversified agroforestry and traceable value chains.

Why this earns marks: directive fidelity, named evidence, analysis, qualification and a reasoned verdict.

2023 GS-I Q17

Give reasons for India's movement from net food importer in the 1960s to net food exporter.

Claim: Explain technology plus institutions: irrigation, HYVs, research, credit, procurement, storage, roads and later diversification.

- **Named evidence/example:** Punjab-Haryana-western UP supplied the early Green Revolution surplus.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** FCI/central-pool procurement converted regional surplus into national food management.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** Later gains in rice, wheat, horticulture, livestock and fisheries broadened output.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.

Qualification: Net export is commodity- and year-specific and does not prove universal nutritional security or sustainable water use.

Verdict: India's transition was an institutional-spatial transformation; future security requires diversified, climate-resilient and resource-aligned productivity.

Why this earns marks: directive fidelity, named evidence, analysis, qualification and a reasoned verdict.

2025 GS-I Q5

What are non-farm primary activities? How are these related to physiographic features in India?

Claim: Define and explicitly exclude manufacturing; cover fishing, forestry, mining and quarrying.

- **Named evidence/example:** Western shelf/upwelling and estuaries shape marine fishing; floodplains/reservoirs shape inland fishery.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** Climate, altitude and access shape Himalayan, central Indian and north-eastern forestry/NTFP systems.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.
- **Named evidence/example:** Peninsular shields, sedimentary basins, lateritic caps and riverbeds govern mining/quarrying occurrence.
- **Analysis:** The evidence proves a specific spatial or institutional mechanism.

Qualification: Physiography sets occurrence and potential, while transport, technology, rights and regulation determine realised use.

Verdict: India's non-farm primary geography is the economic expression of its geological, fluvial, coastal and ecological diversity.

Why this earns marks: directive fidelity, named evidence, analysis, qualification and a reasoned verdict.

Verified cross-topic Mains PYQs

2018 GS-III Q3 - 10 marks

Discuss how MSP can protect farmers from low-income traps.

Claim: MSP can reduce downside price risk only when the signal is credible at the farmer's market.

- **Named evidence/example:** CACP recommendation and CCEA announcement establish the national signal. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FCI/state procurement in rice-wheat belts shows how purchase makes the signal effective. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Pulses/oilseeds show weaker and more episodic procurement geography. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: MSP does not address yield loss, rising cost, tiny marketed surplus or non-farm household risk.

Verdict: Use crop- and region-specific procurement with storage, diversification and risk instruments rather than treating MSP as universal income policy.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2018 GS-III Q4 - 10 marks

Examine the role of supermarkets in agricultural supply chains and whether they eliminate intermediaries.

Claim: Agricultural-marketing performance depends on functions, competition and infrastructure rather than the legal identity of the intermediary.

- **Named evidence/example:** e-NAM demonstrates digital price discovery but still requires assaying, logistics and settlement. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs aggregate small lots and bargaining power but need working capital and governance. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Warehouse receipts and cold chains reduce distress sale and quality loss when accredited and connected. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A large private buyer can replace a commission agent yet reproduce monopsony and quality-dispute risk.

Verdict: Reform should combine market choice, trusted grading, farmer aggregation, finance, logistics and enforceable contracts.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2018 GS-III Q8 - 10 marks

Discuss ecological and economic benefits of Sikkim as an organic state.

Claim: Ecological farming can reduce external input and soil-water pressure while creating differentiated transition costs.

- **Named evidence/example:** Sikkim provides a named whole-state organic-policy example. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** NPOP/PGS-type certification distinguishes standard-based organic from generic low-input practice. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Diversified rotations, manure and biological processes can improve soil functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Certification, market premium and lower chemical use do not guarantee yield, income or zero externality in every crop and season.

Verdict: Phase transitions with extension, local nutrient plans, markets and outcome monitoring.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2018 GS-III Q13 - 15 marks

Assess the role of the National Horticulture Mission in production and farmer income.

Claim: High-value horticulture can raise value per hectare and employment only when perishability and market risk are managed.

- **Named evidence/example:** Mission-linked nurseries, planting material and cluster support strengthen the production base. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Packhouses, cold chains and processing convert biological output into saleable quality. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs and contracts can aggregate produce and connect buyers. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A high gross value can coexist with high cost, rejection, price crash and post-harvest loss.

Verdict: Choose crops by agro-climate, water, market, logistics and household risk rather than price expectation alone.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2018 GS-III Q14 - 15 marks

Elaborate changes in cropping pattern and the emphasis on millet production.

Claim: Cropping patterns respond to agro-climate and relative expected returns shaped by procurement, consumption, technology and value chains.

- **Named evidence/example:** Rice-wheat procurement and irrigation created path dependence in the north-west. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Millets offer dryland, nutrition and climate advantages in suitable regions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Pulses/oilseeds and horticulture require seed, processing, storage and market support. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Diversification can shift water or price risk rather than reduce it when alternative chains are weak.

Verdict: Align price, procurement, water, processing and insurance signals with regional agro-ecology.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2019 GS-III Q3 - 10 marks

Discuss how Integrated Farming Systems sustain agricultural production.

Claim: Integrated Farming Systems stabilise smallholder livelihoods by linking crop, livestock, fish and trees through material and income flows.

- **Named evidence/example:** Crop residues feed livestock and manure returns nutrients. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Farm ponds can support protective irrigation and fish where ecologically suitable. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Multiple enterprises spread seasonal labour and market risk. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Management, labour, animal health, water and market complexity can exceed smallholder capacity.

Verdict: Promote locally designed enterprise combinations with extension, biosecurity and FPO/value-chain support.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2019 GS-III Q4 - 10 marks

Elaborate the impact of watershed development on water-stressed agriculture.

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2019 GS-III Q5 - 10 marks

Discuss the contributions of M. Visvesvaraya and M.S. Swaminathan to water engineering and agricultural science.

Claim: Agricultural transformation emerged from complementary engineering, crop science and state institutions.

- **Named evidence/example:** Visvesvaraya's automatic sluice gates and block irrigation illustrate systematic water engineering. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Swaminathan's role in adapting semi-dwarf wheat/rice links global germplasm to Indian agronomy. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Punjab-Haryana-western UP show the package of water, fertiliser, credit, extension and procurement. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Success was regionally selective and generated groundwater, nutrient and crop-lock-in costs.

Verdict: Retain research and state capacity while shifting to diversified, resource-aligned and rainfed productivity.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2019 GS-III Q13 - 15 marks

Discuss reforms needed to make foodgrain distribution effective.

Claim: Food security requires availability, access, utilisation and stability; procurement and PDS are linked but distinct institutions.

- **Named evidence/example:** FCI/state procurement converts surplus into central-pool stocks. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** NFSA/TPDS creates a legal entitlement channel. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** ONORC/digitisation improve portability and auditability. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Buffer stocks support emergencies and price management. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Cereal access cannot alone solve nutrition, exclusion, storage loss or ecological production costs.

Verdict: Improve dynamic coverage, grievance routes, storage, diversified baskets and transparent stock-release rules.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2019 GS-III Q14 - 15 marks

Elaborate policy responses to food-processing-sector challenges.

Claim: Food processing connects farm geography to consumer markets by extending shelf life, standardising quality and creating derived demand.

- **Named evidence/example:** Milk cooperatives show aggregation-processing-cold-chain integration. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** Fruit/vegetable packhouses and cold chains reduce quality decay. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** AIF and MoFPI-type infrastructure support can crowd in storage and processing. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Processing can concentrate buyer power, water use, waste and low-quality jobs if regulation and local linkages are weak.

Verdict: Build clusters around resource suitability, farmer organisations, logistics, skills, quality and circular waste use.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2020 GS-III Q3 - 10 marks

What are the constraints in transport and marketing of agricultural produce?

Claim: Agricultural-marketing performance depends on functions, competition and infrastructure rather than the legal identity of the intermediary.

- **Named evidence/example:** e-NAM demonstrates digital price discovery but still requires assaying, logistics and settlement. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs aggregate small lots and bargaining power but need working capital and governance. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Warehouse receipts and cold chains reduce distress sale and quality loss when accredited and connected. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A large private buyer can replace a commission agent yet reproduce monopsony and quality-dispute risk.

Verdict: Reform should combine market choice, trusted grading, farmer aggregation, finance, logistics and enforceable contracts.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2020 GS-III Q4 - 10 marks

What are the challenges and opportunities of food processing for raising farmer income?

Claim: Food processing connects farm geography to consumer markets by extending shelf life, standardising quality and creating derived demand.

- **Named evidence/example:** Milk cooperatives show aggregation-processing-cold-chain integration. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Fruit/vegetable packhouses and cold chains reduce quality decay. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** AIF and MoFPI-type infrastructure support can crowd in storage and processing. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Processing can concentrate buyer power, water use, waste and low-quality jobs if regulation and local linkages are weak.

Verdict: Build clusters around resource suitability, farmer organisations, logistics, skills, quality and circular waste use.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2020 GS-III Q6 - 10 marks

How has agricultural technology changed farming and everyday life?

Claim: Digital agriculture improves observation, coordination and transaction only when data are accurate and services remain accessible and contestable.

- **Named evidence/example:** Digital Agriculture Mission provides the umbrella. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** AgriStack federates farmer, map and crop-sown information with states. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Krishi-DSS and remote sensing support monitoring. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** e-NAM, digital claims and advisories connect production to markets/risk. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Land-linked registries may exclude tenants and women; model inference can be wrong; connectivity and correction matter.

Verdict: Pair digital public infrastructure with assisted access, privacy, ground truth, reasons and appeal.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2020 GS-III Q8 - 10 marks

Explain water-conservation and security features of Jal Shakti Abhiyan.

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2020 GS-III Q13 - 15 marks

What explains the success of the rice-wheat system, and what negative consequences followed?

Claim: Agricultural transformation emerged from complementary engineering, crop science and state institutions.

- **Named evidence/example:** Visvesvaraya's automatic sluice gates and block irrigation illustrate systematic water engineering. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Swaminathan's role in adapting semi-dwarf wheat/rice links global germplasm to Indian agronomy. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Punjab-Haryana-western UP show the package of water, fertiliser, credit, extension and procurement. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Success was regionally selective and generated groundwater, nutrient and crop-lock-in costs.

Verdict: Retain research and state capacity while shifting to diversified, resource-aligned and rainfed productivity.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2020 GS-III Q14 - 15 marks

Suggest measures to improve water storage and irrigation under depletion.

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2021 GS-III Q3 - 10 marks

How did land reforms affect marginal and small farmers?

Claim: Land reform succeeds when legal design, records, mobilisation and administration alter the cultivator's actual rights and incentives.

- **Named evidence/example:** Operation Barga recorded sharecroppers. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Kerala and early J&K illustrate stronger reform under distinct political conditions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Agriculture Census separates operated from owned area. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Consolidation and clear leasing can improve investment and scale. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Ceiling evasion, exemptions, informal tenancy and fragmentation limited impact; one state model is not universally transferable.

Verdict: Complete the rights-records-leasing-consolidation chain while protecting tenants, women and smallholders.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2021 GS-III Q4 - 10 marks

How far can micro-irrigation solve India's water crisis?

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2021 GS-III Q13 - 15 marks

What are the salient features of NFSA 2013, and how has it addressed hunger?

Claim: Food security requires availability, access, utilisation and stability; procurement and PDS are linked but distinct institutions.

- **Named evidence/example:** FCI/state procurement converts surplus into central-pool stocks. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** NFSA/TPDS creates a legal entitlement channel. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** ONORC/digitisation improve portability and auditability. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Buffer stocks support emergencies and price management. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Cereal access cannot alone solve nutrition, exclusion, storage loss or ecological production costs.

Verdict: Improve dynamic coverage, grievance routes, storage, diversified baskets and transparent stock-release rules.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2021 GS-III Q14 - 15 marks

What are the challenges of crop diversification and opportunities from emerging technologies?

Claim: Cropping patterns respond to agro-climate and relative expected returns shaped by procurement, consumption, technology and value chains.

- **Named evidence/example:** Rice-wheat procurement and irrigation created path dependence in the north-west. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Millets offer dryland, nutrition and climate advantages in suitable regions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Pulses/oilseeds and horticulture require seed, processing, storage and market support. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Diversification can shift water or price risk rather than reduce it when alternative chains are weak.

Verdict: Align price, procurement, water, processing and insurance signals with regional agro-ecology.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2022 GS-III Q3 - 10 marks

Discuss PDS challenges and measures for effectiveness and transparency.

Claim: Food security requires availability, access, utilisation and stability; procurement and PDS are linked but distinct institutions.

- **Named evidence/example:** FCI/state procurement converts surplus into central-pool stocks. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** NFSA/TPDS creates a legal entitlement channel. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** ONORC/digitisation improve portability and auditability. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Buffer stocks support emergencies and price management. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Cereal access cannot alone solve nutrition, exclusion, storage loss or ecological production costs.

Verdict: Improve dynamic coverage, grievance routes, storage, diversified baskets and transparent stock-release rules.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2022 GS-III Q4 - 10 marks

Elaborate the scope and significance of food processing in India.

Claim: Food processing connects farm geography to consumer markets by extending shelf life, standardising quality and creating derived demand.

- **Named evidence/example:** Milk cooperatives show aggregation-processing-cold-chain integration. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Fruit/vegetable packhouses and cold chains reduce quality decay. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** AIF and MoFPI-type infrastructure support can crowd in storage and processing. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Processing can concentrate buyer power, water use, waste and low-quality jobs if regulation and local linkages are weak.

Verdict: Build clusters around resource suitability, farmer organisations, logistics, skills, quality and circular waste use.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2022 GS-III Q13 - 15 marks

Discuss upstream and downstream bottlenecks in agricultural marketing.

Claim: Agricultural-marketing performance depends on functions, competition and infrastructure rather than the legal identity of the intermediary.

- **Named evidence/example:** e-NAM demonstrates digital price discovery but still requires assaying, logistics and settlement. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs aggregate small lots and bargaining power but need working capital and governance. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Warehouse receipts and cold chains reduce distress sale and quality loss when accredited and connected. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A large private buyer can replace a commission agent yet reproduce monopsony and quality-dispute risk.

Verdict: Reform should combine market choice, trusted grading, farmer aggregation, finance, logistics and enforceable contracts.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2022 GS-III Q14 - 15 marks

Explain benefits of Integrated Farming Systems for small and marginal farmers.

Claim: Integrated Farming Systems stabilise smallholder livelihoods by linking crop, livestock, fish and trees through material and income flows.

- **Named evidence/example:** Crop residues feed livestock and manure returns nutrients. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Farm ponds can support protective irrigation and fish where ecologically suitable. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Multiple enterprises spread seasonal labour and market risk. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Management, labour, animal health, water and market complexity can exceed smallholder capacity.

Verdict: Promote locally designed enterprise combinations with extension, biosecurity and FPO/value-chain support.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2023 GS-III Q3 - 10 marks

Explain how e-technology helps farmers in production and marketing.

Claim: Digital agriculture improves observation, coordination and transaction only when data are accurate and services remain accessible and contestable.

- **Named evidence/example:** Digital Agriculture Mission provides the umbrella. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** AgriStack federates farmer, map and crop-sown information with states. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Krishi-DSS and remote sensing support monitoring. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** e-NAM, digital claims and advisories connect production to markets/risk. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Land-linked registries may exclude tenants and women; model inference can be wrong; connectivity and correction matter.

Verdict: Pair digital public infrastructure with assisted access, privacy, ground truth, reasons and appeal.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2023 GS-III Q4 - 10 marks

Discuss land-reform objectives, measures and land-ceiling policy.

Claim: Land reform succeeds when legal design, records, mobilisation and administration alter the cultivator's actual rights and incentives.

- **Named evidence/example:** Operation Barga recorded sharecroppers. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Kerala and early J&K illustrate stronger reform under distinct political conditions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Agriculture Census separates operated from owned area. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Consolidation and clear leasing can improve investment and scale. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Ceiling evasion, exemptions, informal tenancy and fragmentation limited impact; one state model is not universally transferable.

Verdict: Complete the rights-records-leasing-consolidation chain while protecting tenants, women and smallholders.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2023 GS-III Q13 - 15 marks

Explain cropping-pattern changes driven by consumption and marketing.

Claim: Cropping patterns respond to agro-climate and relative expected returns shaped by procurement, consumption, technology and value chains.

- **Named evidence/example:** Rice-wheat procurement and irrigation created path dependence in the north-west. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Millets offer dryland, nutrition and climate advantages in suitable regions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Pulses/oilseeds and horticulture require seed, processing, storage and market support. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Diversification can shift water or price risk rather than reduce it when alternative chains are weak.

Verdict: Align price, procurement, water, processing and insurance signals with regional agro-ecology.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2023 GS-III Q14 - 15 marks

Discuss Indian farm subsidies and WTO disputes on agricultural support.

Claim: Farm support must be classified by design and trade effect rather than by domestic policy label.

- **Named evidence/example:** WTO amber box captures trade-distorting support subject to commitments. **Analysis:** it demonstrates the causal or spatial relationship demanded.

- **Named evidence/example:** Green box requires specified non/minimally trade-distorting design. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Blue box links support to production-limiting programmes. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Public stockholding debates show the interaction of administered prices and food security. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A measure can serve legitimate livelihood/food-security goals and still face notification or calculation disputes.

Verdict: Improve transparent notification, targeted decoupled support and a durable food-security solution while protecting policy space.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2024 GS-III Q3 - 10 marks

Elaborate factors behind successful land reforms in parts of India.

Claim: Land reform succeeds when legal design, records, mobilisation and administration alter the cultivator's actual rights and incentives.

- **Named evidence/example:** Operation Barga recorded sharecroppers. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Kerala and early J&K illustrate stronger reform under distinct political conditions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Agriculture Census separates operated from owned area. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Consolidation and clear leasing can improve investment and scale. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Ceiling evasion, exemptions, informal tenancy and fragmentation limited impact; one state model is not universally transferable.

Verdict: Complete the rights-records-leasing-consolidation chain while protecting tenants, women and smallholders.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2024 GS-III Q4 - 10 marks

Explain the role of millets in health and nutritional security.

Claim: Cropping patterns respond to agro-climate and relative expected returns shaped by procurement, consumption, technology and value chains.

- **Named evidence/example:** Rice-wheat procurement and irrigation created path dependence in the north-west. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Millets offer dryland, nutrition and climate advantages in suitable regions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Pulses/oilseeds and horticulture require seed, processing, storage and market support. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Diversification can shift water or price risk rather than reduce it when alternative chains are weak.

Verdict: Align price, procurement, water, processing and insurance signals with regional agro-ecology.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2024 GS-III Q13 - 15 marks

State challenges of the Indian irrigation system and government measures.

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2024 GS-III Q14 - 15 marks

Elucidate the importance of buffer stocks for price stabilisation and storage challenges.

Claim: Food security requires availability, access, utilisation and stability; procurement and PDS are linked but distinct institutions.

- **Named evidence/example:** FCI/state procurement converts surplus into central-pool stocks. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** NFSA/TPDS creates a legal entitlement channel. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** ONORC/digitisation improve portability and auditability. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Buffer stocks support emergencies and price management. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Cereal access cannot alone solve nutrition, exclusion, storage loss or ecological production costs.

Verdict: Improve dynamic coverage, grievance routes, storage, diversified baskets and transparent stock-release rules.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2025 GS-III Q3 - 10 marks

Explain factors influencing farmers' selection of high-value crops.

Claim: High-value horticulture can raise value per hectare and employment only when perishability and market risk are managed.

- **Named evidence/example:** Mission-linked nurseries, planting material and cluster support strengthen the production base. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Packhouses, cold chains and processing convert biological output into saleable quality. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs and contracts can aggregate produce and connect buyers. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A high gross value can coexist with high cost, rejection, price crash and post-harvest loss.

Verdict: Choose crops by agro-climate, water, market, logistics and household risk rather than price expectation alone.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2025 GS-III Q4 - 10 marks

Elaborate the scope and significance of supply-chain management of agricultural commodities.

Claim: Agricultural-marketing performance depends on functions, competition and infrastructure rather than the legal identity of the intermediary.

- **Named evidence/example:** e-NAM demonstrates digital price discovery but still requires assaying, logistics and settlement. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** FPOs aggregate small lots and bargaining power but need working capital and governance. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Warehouse receipts and cold chains reduce distress sale and quality loss when accredited and connected. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: A large private buyer can replace a commission agent yet reproduce monopsony and quality-dispute risk.

Verdict: Reform should combine market choice, trusted grading, farmer aggregation, finance, logistics and enforceable contracts.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 10-marker.

2025 GS-III Q13 - 15 marks

Examine factors behind groundwater depletion and government steps.

Claim: India's agricultural water crisis is a source-distribution-application-drainage-governance problem, not only a shortage of projects.

- **Named evidence/example:** PMKSY separates access, efficiency and watershed/command functions. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Drip/sprinkler can raise plot efficiency; basin saving depends on rebound. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Watershed ridge-to-valley treatment supports recharge and protective irrigation. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** The groundwater-power nexus in north-western and hard-rock regions shows shared-stock externality. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Potential created or area covered does not prove reliable, equitable and sustainable irrigation.

Verdict: Combine aquifer budgeting, crop-water alignment, energy reform, canal maintenance, drainage and participatory allocation.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

2025 GS-III Q14 - 15 marks

Examine the scope of food processing and its employment potential.

Claim: Food processing connects farm geography to consumer markets by extending shelf life, standardising quality and creating derived demand.

- **Named evidence/example:** Milk cooperatives show aggregation-processing-cold-chain integration. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** Fruit/vegetable packhouses and cold chains reduce quality decay. **Analysis:** it demonstrates the causal or spatial relationship demanded.
- **Named evidence/example:** AIF and MoFPI-type infrastructure support can crowd in storage and processing. **Analysis:** it demonstrates the causal or spatial relationship demanded.

Qualification: Processing can concentrate buyer power, water use, waste and low-quality jobs if regulation and local linkages are weak.

Verdict: Build clusters around resource suitability, farmer organisations, logistics, skills, quality and circular waste use.

Why this earns marks: mark-proportionate evidence, explicit mechanism, counterpoint and verdict for a 15-marker.

Original 10-mark practice 1

Question: Examine the continuing relevance of Von Thunen for peri-urban agriculture in India.

Claim: Von Thunen remains relevant as a mechanism of transport cost, perishability and land rent, even though Indian cities no longer display perfect concentric rings.

- **Named evidence/example:** Milk, leafy vegetables and flowers cluster around large markets such as Delhi and Bengaluru. **Analysis:** frequent delivery and quality loss make access valuable, so high-value perishables can outbid extensive uses near demand. **Qualification:** urban expansion can displace these belts outward.
- **Named evidence/example:** Expressways, refrigerated transport and collection centres extend the distance from which perishables can arrive. **Analysis:** technology stretches the ring by reducing decay and travel time. **Qualification:** energy, coordination and cold-chain reliability add new costs.
- **Named evidence/example:** Competing wholesale markets and processors create several demand nodes rather than one isolated market. **Analysis:** the rent gradient becomes corridor- and network-shaped. **Qualification:** this departs from the model's single-market assumption.

Verdict: The classical map is obsolete as a literal pattern, but its distance-cost logic remains a strong first explanation of peri-urban high-value farming.

Why this earns marks: It answers “examine,” applies the model to named Indian urban contexts, explains modern modification, states assumptions and gives a graded verdict.

Original 10-mark practice 2

Question: Explain why an increase in irrigation potential need not produce reliable and sustainable irrigation.

Claim: Irrigation potential measures created capacity; reliable and sustainable irrigation depends on actual delivery, equitable access, drainage, aquifer balance and crop-water use.

- **Named evidence/example:** Canal commands may suffer poor maintenance, tail-end deprivation and waterlogging. **Analysis:** created capacity does not prove timely field delivery. **Qualification:** well-managed commands can still provide stable multi-season irrigation.
- **Named evidence/example:** Punjab's groundwater-energy nexus lowers the marginal pumping cost. **Analysis:** individual pumping from a shared aquifer can produce collective depletion. **Qualification:** groundwater can remain a flexible buffer where recharge and abstraction are balanced.
- **Named evidence/example:** Hard-rock aquifers in the Deccan have limited and uneven storage. **Analysis:** more wells can divide a small stock rather than create more dependable water. **Qualification:** watershed recharge and aquifer mapping can improve local resilience.
- **Named evidence/example:** PMKSY distinguishes access, efficiency and watershed or command functions. **Analysis:** the policy design itself shows that infrastructure, application efficiency and source sustainability are different problems. **Qualification:** programme coverage remains an activity measure, not proof of basin-level savings.

Verdict: Irrigation becomes reliable only when source, distribution, application, drainage and governance are managed as one hydrological system.

Why this earns marks: It distinguishes potential from outcome, uses canal and groundwater mechanisms, adds a programme example and ends with a systems verdict.

Original 15-mark practice 1

Question: Analyse how land tenure, water, procurement and cold-chain geography jointly shape crop choice in India.

Claim: Crop choice is a household risk decision shaped by control over land, water reliability, assured sale and the ability to preserve quality-not by agro-climate or price alone.

- **Named evidence/example:** The 2015-16 Agriculture Census records 146.45 million operational

holdings with an average size of 1.08 hectares. **Analysis:** small operational scale can reduce risk-bearing capacity and favour familiar crops with established buyers. **Qualification:** aggregation and custom services can partly overcome scale constraints.

- **Named evidence/example:** Irrigated north-western districts support rice-wheat rotations.

Analysis: assured water permits timely sowing and lowers yield risk. **Qualification:** the same choice can become ecologically irrational where aquifers are stressed.

- **Named evidence/example:** FCI and state procurement concentrate credible purchase in

selected grain belts. **Analysis:** a visible buyer converts MSP from a signal into a bankable expectation. **Qualification:** MSP announcement does not mean equal procurement across crops and states.

- **Named evidence/example:** Horticultural belts such as Nashik depend on packhouses, transport,

storage and processing. **Analysis:** cold-chain access makes a perishable high-value crop saleable beyond the local market. **Qualification:** rejection, power cost and buyer concentration can replace biological risk with market risk.

- **Named evidence/example:** FPOs and e-NAM-type systems can aggregate lots and improve price

discovery. **Analysis:** institutions can widen crop options for small operators. **Qualification:** registration or platform value is not proof of higher net farm income.

Verdict: Diversification succeeds when tenure security, water, purchase assurance and perishability management are aligned regionally; changing only the announced price is insufficient.

Why this earns marks: It analyses four interacting determinants, uses dated structural evidence, adds regional examples and qualifies every policy mechanism.

Original 15-mark practice 2

Question: Critically examine the regional diffusion and second-generation challenges of the Green Revolution.

Claim: The Green Revolution was a technology-institution package that secured national grain supply but diffused unevenly because its preconditions were geographically selective.

- **Named evidence/example:** Punjab, Haryana and western Uttar Pradesh combined assured

irrigation, level alluvial land, credit, extension and procurement. **Analysis:** complementary preconditions produced an early wheat-rice surplus core. **Qualification:** technology alone cannot explain the concentration.

- **Named evidence/example:** FCI and state procurement connected this regional surplus to the

central pool. **Analysis:** market assurance made repeated input-intensive cultivation rational. **Qualification:** procurement geography remained narrower than production geography.

- **Named evidence/example:** Later diffusion into parts of Rajasthan, Madhya Pradesh and other

irrigated districts shows that the package could travel where conditions improved. **Analysis:** diffusion followed infrastructure rather than a uniform national wave. **Qualification:** many rainfed and tribal regions remained outside the strongest gains.

- **Named evidence/example:** Rice-wheat repetition in the north-west increased groundwater,

nutrient and residue-management pressures. **Analysis:** the successful core acquired second-generation ecological costs. **Qualification:** these costs arise from incentive and resource misalignment, not from yield improvement in itself.

- **Named evidence/example:** Pulses, oilseeds and millets often remained in less-favoured

dryland areas. **Analysis:** crop-pattern narrowing shifted diversity and risk spatially. **Qualification:** diversification without processing and purchase support can expose farmers to new price risk.

Verdict: The next transformation should retain research, extension and state capacity while redirecting them toward rainfed productivity, crop-water alignment, soil recovery and credible markets for diversified crops.

Why this earns marks: It evaluates both achievement and uneven diffusion, ties evidence to spatial mechanisms, avoids a one-sided ecological critique and offers mechanism-matched reform.

Original 20-mark practice 1

Question: Agricultural transformation must move from output maximisation to resilient value creation. Discuss with reference to rainfed systems, markets, digital agriculture and a just transition.

Claim: Resilient value creation means raising stable net livelihood value per unit of scarce water, soil and risk, rather than maximising a single crop's gross output.

- **Named evidence/example:** Watershed ridge-to-valley treatment and farm ponds in rainfed

regions support recharge and protective irrigation. **Analysis:** small, timely water access can stabilise crops without copying canal-intensive systems. **Qualification:** structures need catchment logic and maintenance; counting structures is not an outcome.

- **Named evidence/example:** Millets, pulses, oilseeds and integrated crop-livestock systems fit

many dryland ecologies. **Analysis:** diversification spreads climatic and income risk while improving biomass and nutrient flows. **Qualification:** alternative crops need seed, processing and dependable demand.

- **Named evidence/example:** FPOs, grading, warehouse finance, cold chains and processing can

convert output into retained value. **Analysis:** aggregation and quality preservation improve bargaining and reduce distress sale. **Qualification:** governance, working capital and monopsony risks remain.

- **Named evidence/example:** The Digital Agriculture Mission was approved on 2 September 2024,

while AgriStack uses federated registries. **Analysis:** interoperable data can support targeted services and decision support. **Qualification:** registry creation is not impact; consent, correction, exclusion and accountability are essential.

- **Named evidence/example:** Sikkim and the National Mission on Natural Farming, approved on

25 November 2024, illustrate ecological-transition pathways. **Analysis:** lower external input dependence may improve soil-biological processes and differentiated market value. **Qualification:** organic, natural and conservation agriculture are distinct; yield and income effects vary by crop and transition stage.

- **Named evidence/example:** Water-stressed rice-wheat regions and vulnerable smallholders show

why transition costs are unequal. **Analysis:** new incentives can create short-run income, skill and asset losses for some groups. **Qualification:** reform must be sequenced rather than imposed as a uniform withdrawal of support.

Verdict: A resilient transformation combines agro-ecological fit, risk-spreading farm systems, competitive value chains, accountable digital public infrastructure and time-bound transition support for affected farmers and workers.

Why this earns marks: It directly defines the requested shift, covers all four named dimensions, uses dated programme evidence, links every example to a mechanism and builds a balanced just-transition verdict.

Original 20-mark practice 2

Question: Evaluate India's food-security geography from production and procurement to stocks, PDS, trade and nutrition.

Claim: India's food security is a spatial chain linking agro-climatic production, selective procurement, central-pool movement and legal distribution; success in cereal availability does not by itself secure nutrition or sustainability.

- **Named evidence/example:** Punjab, Haryana and western Uttar Pradesh formed an early

Green-Revolution surplus core. **Analysis:** irrigation and technology converted a regional advantage into national grain availability. **Qualification:** concentration also created groundwater and crop-lock-in costs.

- **Named evidence/example:** FCI and state agencies procure in surplus belts and move grain

into the central pool. **Analysis:** procurement and logistics translate local surplus into inter-regional redistribution. **Qualification:** procurement is more concentrated than production and varies by crop.

- **Named evidence/example:** NFSA and TPDS create an entitlement channel, while portability initiatives help mobile households use it beyond the home location. **Analysis:** legal and digital architecture addresses access, not merely availability. **Qualification:** exclusion, authentication and grievance failures can still interrupt access.

- **Named evidence/example:** Buffer stocks support emergencies, distribution and market

management. **Analysis:** stocks create temporal stability between harvest and need. **Qualification:** stock level, storage quality and release rules matter; production is not the same as stock.

- **Named evidence/example:** Trade can bridge commodity or seasonal gaps and dispose of

selected surpluses. **Analysis:** it adds flexibility to a geographically uneven system. **Qualification:** net export is commodity- and year-specific and cannot prove universal food security.

- **Named evidence/example:** Cereal entitlements coexist with nutrition challenges requiring

pulses, oils, diverse foods, health, water and sanitation. **Analysis:** utilisation and diet quality extend beyond grain calories. **Qualification:** a diversified basket must remain logistically and fiscally workable.

- **Named evidence/example:** The v1 data firewall separates Third Advance Estimates,

procurement, stocks, distribution, consumption and export. **Analysis:** correct labels prevent false causal claims. **Qualification:** even accurate national aggregates can hide regional and household inequality.

Verdict: India should preserve the production-procurement-PDS backbone while diversifying crops and baskets, improving dynamic coverage and grievance systems, modernising storage and aligning production incentives with regional water and nutrition needs.

Why this earns marks: It evaluates the full chain named in the question, distinguishes every data category, integrates geography with institutions and nutrition, and ends with a source-aware graded verdict.